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SGS 501/601 RESEARCH METHODS / RESEARCH METHODS and SCIENTIFIC ETHICS



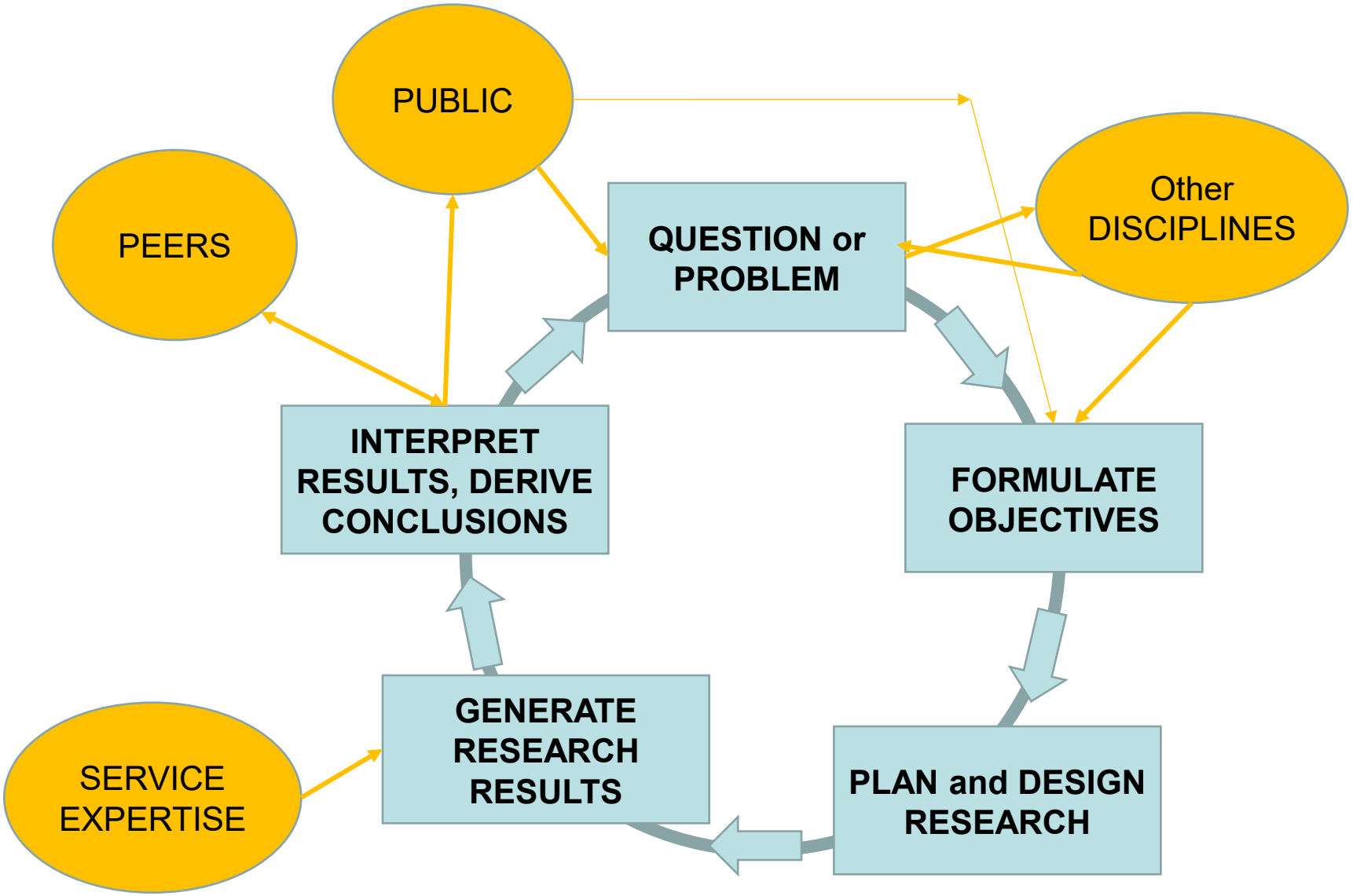
LITERATURE REVIEW

AGENDA

- Steps of Research Process and Plan reviewed
- Why to conduct Literature Review?
- How to conduct Literature Review?
- How to write Literature Review?

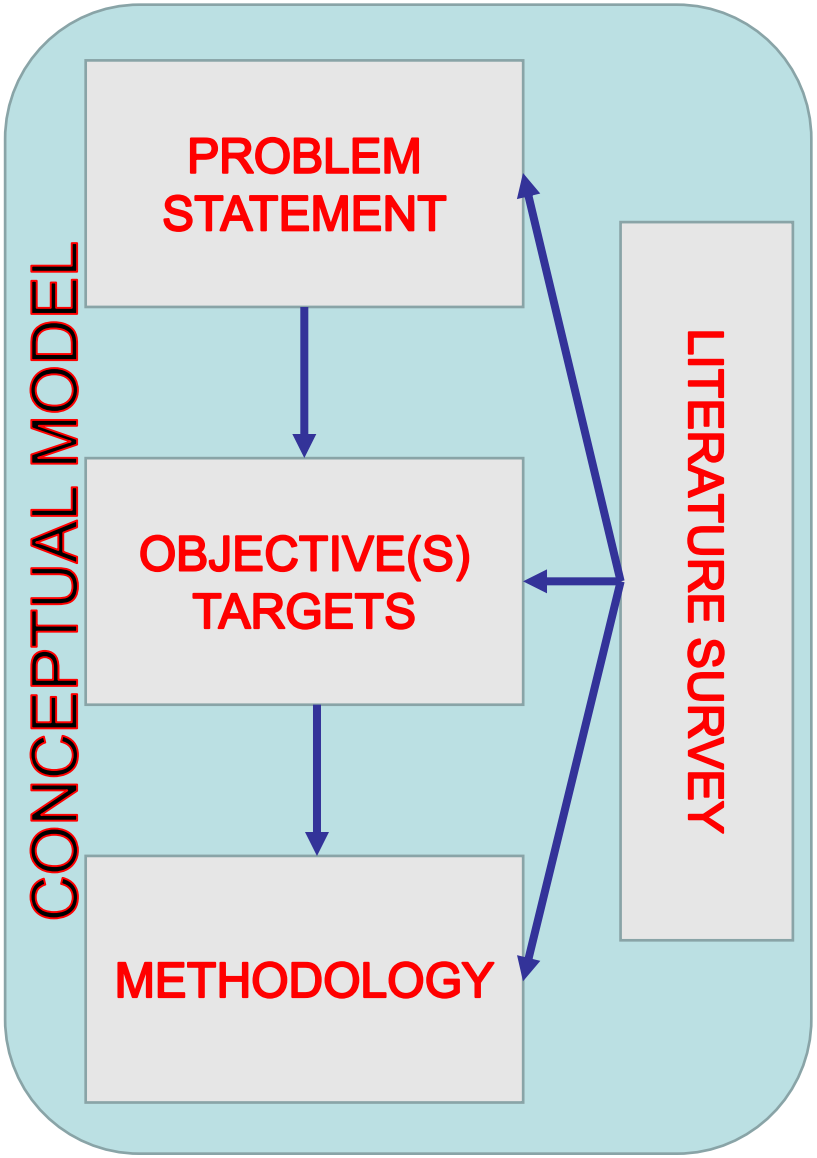
Subjective opinion of the instructor:
You will find 600 (2000) papers in your searches,
read abstract of 300 (1000) papers,
download 150 (500) papers,
fully read 25 (100) papers,
include 75 (150) papers in the bibliography
for Master's (PhD) thesis.

STEPS of the RESEARCH PROCESS



Where to start: Make a plan!

- Describe the problem
- Identify objectives
- Specify methodology
- Develop conceptual model
- Literature survey:
  - to provide research problem
  - to provide base knowledge on problem
  - to provide methods to be used
- DOCUMENTATION



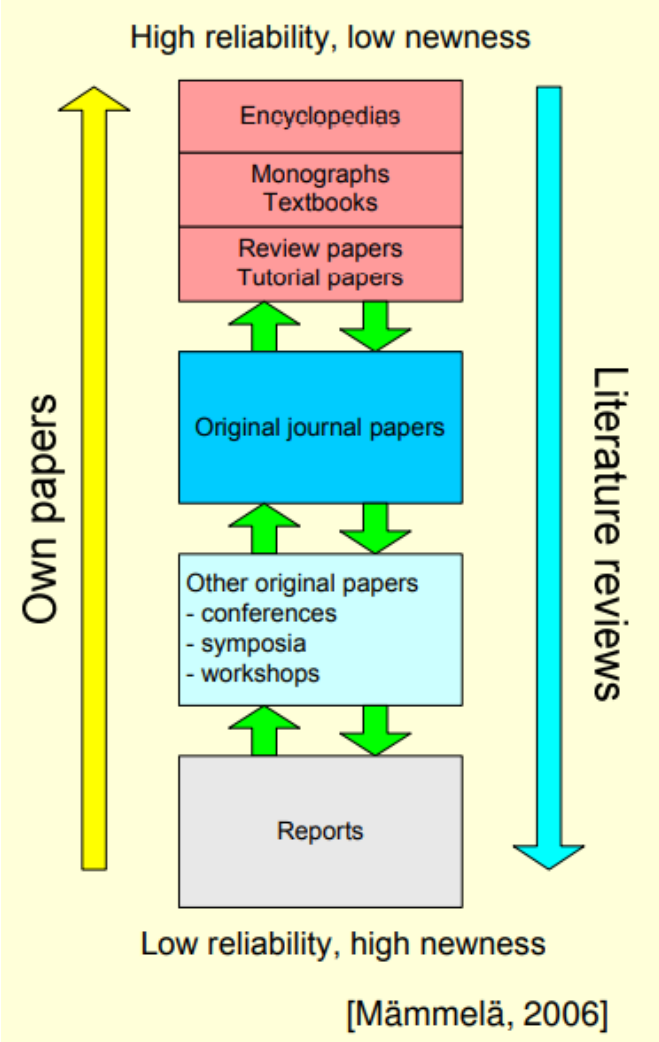
Purpose of Literature Review

The function of a scientific literature review is primarily to collect and examine the state of current knowledge in a field by examining the work of scholars and researchers whose work has been recognized as valuable.

1. Establishes context for your work by showing what has been done in the area including strengths and weaknesses
2. Exposes the gap in current knowledge with your research problem in focus
3. Shows your supervisors that you have done your research!

Use of Literature Review

- How has the work been done previously? Which methods and procedures have been used?
- What similar work has been leading up to this point?
- What distinguishes previous work from what you want to do?
- Who / What will be impacted by this research?



Benefits of Literature Review

A. ***Prevents duplication of what has already been done*** (Some duplication or confirmation of research is necessary, but excessive duplication is wasteful)

B. ***Help to identify new areas where research is needed*** (and how new research can contribute)

C. ***Provides ideas and direction for:***
   1. How to handle problems encountered
   2. Techniques
   3. Sources of data
   4. Novel approaches for the research

D. ***Helps develop insights on design of your own study by showing what has (and has not) been previously successful***

E. ***May reveal conceptual insights into the problem and/or suggest possible hypotheses for your own study***

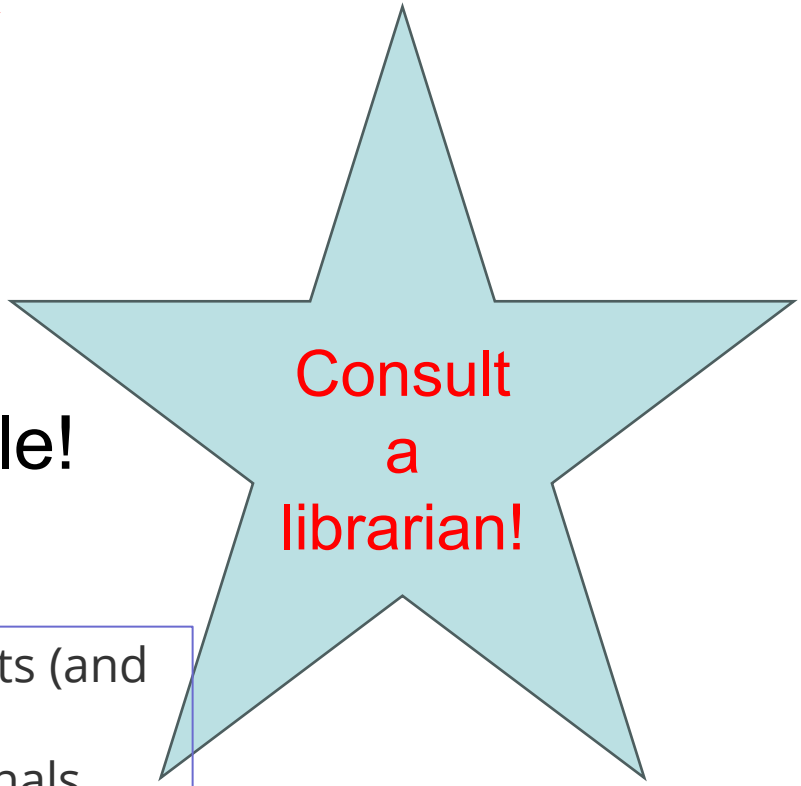
How to Perform Literature Review?

- Include "scientific literature": *professional journals, formal research reports, university-affiliated bulletins, reports and monographs (including theses) passed through "peer review"* (No guarantee that these are infallible, or even correct – but they have been through independent checks of accuracy and correctness.)
- Use a research library: bilgimerkezi.khas.edu.tr
- Use search tools including indexing services such as Web Of Science and Scopus

  bilgimerkezi.khas.edu.tr: Research Tools -> Databases
- Use "key words": start with a broad list of key words, then narrow to a confined list. Hint: look at key words of "review or tutorial papers" related to your topic.
- Start reading most recent publications => focus on current knowledge, they include references to earlier research!
- Read abstract first to decide: to include or not to include!
- Take notes (preferably digitally in an organized manner!): objective, methods, findings, conclusion, "shortcomings"

Kütüphane-Library

- 7/24 online access
- Nice place with nice people!



Web of Science: Citations with abstracts (and full texts when available) to scholarly literature in all areas of research (journals, books, proceedings of conferences, etc.)

Databases: SCI, SCIE (Science Citation Index Expanded), ESCI (Emerging Science Citation Index), Social Sciences Citation Index, Arts & Humanities Citation Index, Conference Proceedings Citation Index

Use a Tool Appropriate for the Format Required

What is Reference Management Software?

Reference management software allows researchers to link to and correctly cite referenced works in their papers. Also known as citation management tools, the software can create proper citations in a desired style (e.g. MLA, APA, etc.). Additionally, reference management software may automate building bibliographies, or footnotes, or reference sections in research papers. Reference management software may organize personal research libraries, or manage access to an outside reference source (e.g. web magazines, literature databases).

From: https://www.trustradius.com/reference-management

Alternatives (sorted alphabetically):

BibTex, Cite4Me, Cite This for Me, EndNote, Mendeley, MyLib, refbase, RefWorks, Zenreader, Zotero,

OBTAIN the template of your target document (paper, report, thesis etc.) and practice with it! (Start typing directly into the template!)

Steps of Literature Review - I

a) Organizing: Develop a framework by identifying the key articles or the key areas of knowledge and associating papers with specific areas of knowledge. This set of topics or papers should form the sections of your literature review; however, you'll need to organize these topics logically, and develop transitions between the sections.

b) Summarizing: Identify each article's contribution to the area of knowledge. You may be summarizing an entire article, or just including a brief reference to the article. When summarizing an article, ask the following questions:

- What is the author's purpose?
- What are the writer's assumptions?
- What are the author's main claims (conclusions)?
- How are they supported; how have they been qualified ?

c) Evaluating: Assess the work done in the key area of knowledge or by the pivotal paper, in order to establish:

- How previous work has left a gap, because of either inadequate assumptions or inconclusive findings;
- How previous research will be applied in a new context; or
- How general disagreement or different views on the subject create a need for a solution

From: https://ecp.engineering.utoronto.ca/resources/online-handbook/components-of-documents/literature-reviews/

Steps of Literature Review - II

In evaluating each article, consider the following questions:

- How strong are the basic components of the study design? Could the problem have been approached more effectively from another perspective or with different assumptions?
- Are the paper's conclusions well warranted by evidence from research? Is the evidence from the research conclusive? Or are there limitations to the research?
- How does this paper contribute to our understanding of the problem/issue?
- How does the paper relate to your research ?
- In evaluating several papers on a specific topic, ask these questions:
- What are the significant points of agreement between articles?
- Where the research disagrees, is one researcher more conclusive than another?
- How can you fit the articles together to build a logical argument that furthers your purpose?

From: https://ecp.engineering.utoronto.ca/resources/online-handbook/components-of-documents/literature-reviews/

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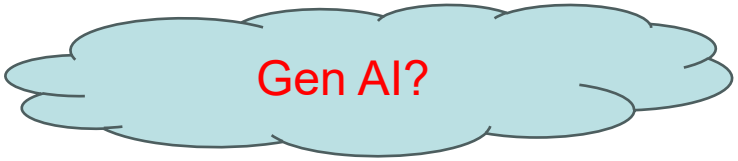
Writing the Literature Review

1. Organize: develop outline, make an introduction, group work with similar focus, end with a short summary

   The review is not just a list of unconnected summaries, but a coherent synthesis of previous literature

   Do not quote directly, do not use figures unless mandatory!

   Do not reference a source of an idea without actually having read it.



2. Referencing: Plagiarism is the failure to give credit for an idea or research result to it's originator. Presenting someone else's words or ideas as your own is not only wrong but can have formal consequences. By properly referencing and giving credit for other's work, you show that you are aware of the state of knowledge in your subject and are familiar with the work of leaders in the field.

Styles for Citation and Referencing

Faculty of Law	**All programs**	**OSCOLA**
Faculty of Communication	All programs	Chicago (Author, Date)
Faculty of Engineering and Natural Sciences		
	Computer Engineering	IEEE
	Bioinformatics and Genetics	Harvard
	Electronic Engineering	IEEE
	Industrial Engineering	APA
	Energy and Sustainable Development	APA
	Finance Engineering	Harvard
	Computational Sciences and Engineering	Harvard
	Computational Biology and Bioinformatics	Harvard
	Materials Science and Nanotechnology	ACS
Faculty of Art and Design	All programs	APA
Faculty of Economics, and Social Sciences	All programs except Master of Business Administration (MBA) in Management (Non-Thesis), Master of Arts in Business Intelligence and Analytics (Non-Thesis) and Psychology	Chicago (Author, Date)
	Master of Business Administration (MBA) in Management (Non-Thesis), Master of Arts in Business Intelligence and Analytics (Non-Thesis) and Psychology	APA
Faculty of Administrative Sciences	All programs except Master of Cyber Security	Chicago (Author, Date)
Faculty of Administrative Sciences	Master of Cyber Security	IEEE

APA for Citation and Referencing

https://apastyle.apa.org/instructional-aids/reference-examples.pdf

- ***Parenthetical citation***: (Kushilevitz & Malkin, 2016)
- ***Narrative citation***: Kushilevitz and Malkin (2016)

Journal Article

Niepel, C., Hausen, J. E., Weber, A. M., & Möller, J. (2025). Understanding mean-level and intraindividual variability in state academic self-concept: The role of students' trait expectancies and values. Journal of Educational Psychology, 117(5), 772–788. https://doi.org/10.1037/ edu0000946

Authored Book

Levenson, H., Jinich, S., Vaz, A., & Rousmaniere, T. (2025). Deliberate practice in emotionally focused couple therapy. American Psychological Association. https://doi.org/10.1037/0000436-000

Edited Book Chapter

Zeleke, W. A., Hughes, T. L., & Drozda, N. (2020). Home–school collaboration to promote mind– body health. In C. Maykel & M. A. Bray (Eds.), Promoting mind–body health in schools: Interventions for mental health professionals (pp. 11–26). American Psychological Association. https://doi.org/10.1037/0000157-002

Proceeding

Kushilevitz, E., & Malkin, T. (Eds.). (2016). Lecture notes in computer science: Vol. 9562. Theory of cryptography. Springer. https://doi.org/10.1007/978-3-662-49096-9

Preprint Article

Latimier, A., Peyre, H., & Ramus, F. (2020). A meta-analytic review of the benefit of spacing out retrieval practice episodes on retention. PsyArXiv. https://psyarxiv.com/kzy7u/

Data Set

O'Donohue, W. (2017). Content analysis of undergraduate psychology textbooks (ICPSR 21600; Version V1) [Data set]. Inter-university Consortium for Political and Social Research. https://doi.org/10.3886/ ICPSR36966.v1

Webpage With a Retrieval Date

Worldometer. (n.d.). Current world population. Retrieved October 17, 2025, from https://www. worldometers.info/world-population/

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Use of Generative AI for Research

Exploring the Role, Benefits, and Best Practices

WHO? WHY? WHEN? WHAT? HOW?

WHO Uses Generative AI for Research?

- - **Researchers**: Academics, scientists, students
- - **Professors & Educators:** Teaching and academic writing
- - **Students:** Literature reviews, summarization, idea generation
- - **Industry Professionals:** Market research, trend analysis

WHY Use Generative AI in Research?

- - Efficiency: Saves time in summarizing and structuring research
- - Data Processing: Analyzes large datasets quickly
- - Idea Generation: Helps brainstorm new perspectives
- - Writing Assistance: Improves academic writing
- - Language Support: Assists non-native researchers
- AI enhances researchers' capabilities!

WHEN to Use Generative AI in Research?

- Before Research:
  - Generating ideas & research questions
  - Conducting literature reviews

- During Research:
  - Summarizing & analyzing data
  - Drafting outlines & structuring content

- After Research:
  - Enhancing readability & checking grammar
  - Formatting & citations

Use AI strategically!

WHAT Are the Best Use Cases? & HOW

- Literature Review: Summarizing and identifying research gaps
- Data Interpretation: Assisting with pattern recognition
- Academic Writing: Improving structure & coherence
- Plagiarism Checking: Ensuring originality
- Presentations & Reports: Refining slides & abstracts

HOW to Use Generative AI Responsibly?

- Verify Information: Cross-check AI-generated content
- Cite Properly: AI cannot be a primary source
- Maintain Critical Thinking: AI suggests, but human judgment is key
- Avoid Over-Reliance: AI should assist, not replace research
- Respect Ethics: Follow university guidelines

Guidelines

Permitted vs. Non-Permitted Uses

- **Appropriate (Assistant Role):** Use AI to discover semantic connections, brainstorm search terms, group articles by theme, build comparison tables, and refine language.
- **Inappropriate (Replacement Role):** Do not let AI evaluate study quality, generate text from unread papers, or synthesize core theoretical arguments. AI cannot replace human critical thinking or accountability.

Key Risks to Manage

- **Hallucinated and Fake References:** Large language models frequently invent nonexistent citations, false data, or misinterpret specific findings. Always read (and download a copy) and verify the original paper yourself before citing it.
- **Biases and Omissions:** AI search engines can inherit publication, linguistic, or geographic biases. They may also fail to catch retracted papers or predatory journals.
- **Privacy and Confidentiality:** Avoid uploading unpublished peer-review manuscripts, proprietary data, or human subject data into public AI platforms unless using approved enterprise/local tools.

Standards for Disclosure and Authorship

- **No AI Authorship:** AI tools cannot be listed as authors because they lack legal status and cannot assume academic responsibility. Only human researchers are accountable.
- **Transparent Reporting:** Clearly declare how and where AI was used (such as in the methodology, acknowledgments, or tool appendices). Specify the tool name, version, date of use, and specific prompts or workflows, following publisher guidelines.

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Conclusion & Discussion

- GenAI is a powerful research tool when used wisely

- Ethical considerations & human oversight are crucial

- How do YOU plan to integrate GenAI into your research?

- The future of research is AI-assisted, but human-led!
- AI is a tool, not a substitute for academic integrity!

Final note

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read abstract of 300 (1000) papers,
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fully digest 25 (100) papers,
include 75 (150) papers in the bibliography for Master's (PhD) thesis.

GOOD LUCK!